

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2003

MATHEMATICS

EXTENSION II

*Time Allowed – 3 Hours
(Plus 5 minutes reading time)*

All questions may be attempted

All questions are of equal value

In every question, show all necessary working

Marks may not be awarded for careless or badly arranged work

**Standard integral tables are provided for your convenience.
Approved silent calculators may be used.**

**The answers to all questions are to be returned in separate bundles
clearly labelled Question 1, Question 2, etc. Each bundle must show your
candidate number.**

JRAHS TRIAL - EXT II 2003

Question 1:

- (a) The complex numbers $z_1 = \frac{a}{1+i}$ and $z_2 = \frac{b}{1+2i}$ where a and b are real, satisfy the condition $z_1 + z_2 = 1$. Find the value of a and b . 3
- (b) The complex number z has modulus r and argument θ where $0 \leq \theta \leq \pi$. Find in terms of r and θ the modulus and arguments of
- (i) z^2 1
- (ii) $\frac{1}{z}$ 1
- (iii) iz 1
- (c) (i) Sketch (without using calculus) the curve $y = \frac{x^2 + 2x - 3}{x - 2}$ clearly showing its intercepts with the coordinate axes and the position of all its asymptotes. 5
- (ii) Find the area bounded by the curve $y = \frac{x^2 + 2x - 3}{x - 2}$ and the x -axis. 4

Question 2: (START A NEW PAGE)

- (a) Evaluate:
- (i) $\int_0^{\frac{\pi}{6}} \cos \theta \sin^3 \theta \, d\theta$. 2
- (ii) $\int_0^3 \frac{\sqrt{x}}{1+x} \, dx$. (Let $u^2 = x$). 3
- (iii) $\int_0^{\frac{\pi}{2}} \frac{1}{5 + 3 \cos \theta} \, d\theta$ 4
- (b) Given that $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$.
- (i) Prove that $I_n + n(n-1)I_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$. 4
- (ii) Evaluate $\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$. 2

Question 3: (START A NEW PAGE)

- (a) Sketch the ellipse $9x^2 + 25y^2 = 225$ clearly showing: 4
- (i) the coordinates of the intercepts with the x and y -axes,
 - (ii) the coordinates of the foci,
 - (iii) the equation of the directrices.
- (b) Prove that the curves $x^2 - y^2 = c^2$ and $xy = c^2$ meet at right angles. 4
- (c) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the ellipse meets the x -axis at the points A and A' . 3
- (i) Prove that the tangent at P has the equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 3
 - (ii) The tangent at P meets the tangents from A and A' at points Q and Q' respectively. Find the coordinates of Q and Q' . 2
 - (iii) Prove that the product $AQ \times A'Q'$ is independent of the position of P . 2

Question 4: (START A NEW PAGE)

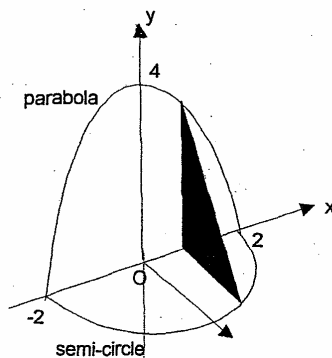
- (a) Prove that $\frac{d}{dx} \left[\sqrt{bx - x^2} + \frac{b}{2} \cos^{-1} \left(\frac{2x - b}{b} \right) \right] = -\sqrt{\frac{x}{b - x}}$ for $x \geq 0$. 3
- (b) A particle of mass m is attracted towards the origin by a force of magnitude $\frac{\mu m}{x^2}$ for $x \neq 0$, where the distance from the origin is x and μ is a positive constant. 3
- (i) If the particle starts from rest at a distance b to the right of the origin, show that its velocity v is given by $v^2 = 2\mu \left(\frac{b - x}{bx} \right)$. 3
 - (ii) Find the time required for the particle to reach a point halfway towards the origin. 4
- (c) Using the Principle of mathematical induction, prove that $(x + 1)^n - nx - 1$ is divisible by x^2 for all integer $n \geq 2$. 5

Question 5: (START A NEW PAGE)

- (a) (i) Using the substitution $x = 2 \sin \theta$, prove that $\int_0^2 (4 - x^2)^{\frac{3}{2}} dx = 16 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$ 2

- (ii) A solid (see diagram) sits on a semi-circular base of radius 2 units. Vertical cross-sections perpendicular to the diameter of the semi-circle are right-angled triangles with their heights being bounded by the parabola $y = 4 - x^2$. By slicing the solid perpendicular to the x -axis, show that the volume (V unit³) of the solid formed is given by

$$V = \int_0^2 (4 - x^2)^{\frac{3}{2}} dx$$



- (iii) Find the volume of the solid. 5

- (b) A tourist is walking along a straight road. At one point he observes a vertical tower standing on a large flat plain. The tower is on a bearing 053° with an angle of elevation of 21° . After walking 230 metres, the tower is on a bearing 342° with an angle of elevation of 26° .

- (i) Draw a neat diagram showing the above information. 1

- (ii) Find the height of the tower correct to the nearest metre. 5

Question 6: (START A NEW PAGE)

- (a) The tangent to the hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ meets the x and y axes at F and G respectively and the normal at T meets the line $y = x$ at H .

- (i) Show that the tangent at T is $x + t^2 y = 2ct$. 3

- (ii) Show that the normal at T is $t^3 x - ty = c(t^4 - 1)$. 2

- (iii) Prove that $FH \perp HG$. 6

- (b) The area bounded by the curve $y = \frac{\ln x}{\sqrt{x}}$ and the x -axis for $1 \leq x \leq e$ is rotated through one revolution about the y -axis. Using the method of cylindrical shells, find the volume of the solid formed. 4

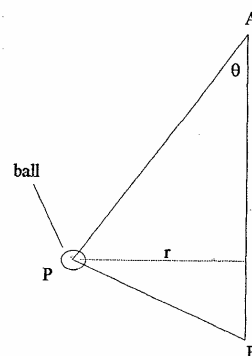
Question 7: (START A NEW PAGE)

- (a) In a state swimming championships, 12 swimmers (including the Jones twins) are chosen to represent their club and are divided into three teams of four swimmers to form 3 relay teams. Find the number of ways this can be done:

- (i) with no restrictions. 2
- (ii) if the Jones twins (Angela and Bethany) are not to be in the same relay team. 3

- (b) The ends of a light string are fixed at 2 points A and B with B directly below A, as shown in the diagram. The string passes through a small ball of mass m which is then fastened to the string at point P. The angle PAB is θ and the distance from P to AB is r .

Suppose now that the ball revolves in a horizontal circle about the vertical through AB with constant angular velocity ω and while this happens both sections (AP and BP) of the string are taut and the angle APB is a right angle.

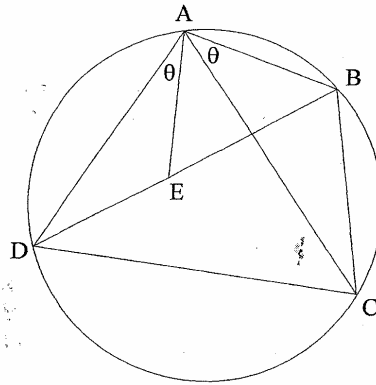


- (i) Draw a diagram showing the forces acting on the ball. 2
- (ii) Show that the tensions T_1 and T_2 in the sections of the string AP and BP respectively are $T_1 = m(r\omega^2 \sin \theta + g \cos \theta)$ and $T_2 = m(r\omega^2 \cos \theta - g \sin \theta)$. 4
- (iii) Given that $AB = 100\text{cm}$ and $AP = 80\text{cm}$, show that $\omega^2 > \frac{25g}{16}$. 2
- (iv) Suppose that the ball is free to slide on the string. Show that the condition for the ball to remain at point P on the string is $\omega^2 = \frac{175g}{12}$. 2

Question 8: (START A NEW PAGE)

- (a) (i) If $t = \tan x$ prove that $\tan 4x = \frac{4t(1-t^2)}{t^4 - 6t^2 + 1}$. 2
- (ii) If $\tan x \tan 4x = 1$ deduce that $5t^4 - 10t^2 + 1 = 0$. 1
- (iii) Prove that $x = 18^\circ$ and $x = 54^\circ$ satisfy the equation $\tan x \tan 4x = 1$. 2
- (iv) Deduce that $\tan 54^\circ = \sqrt{\frac{5+2\sqrt{5}}{5}}$. 3

- (b) ABCD is a cyclic quadrilateral and E is on BD such that $\angle DAE = \angle BAC$.



- (i) Copy the diagram onto your answer sheet and prove that $\triangle ABE$ and $\triangle ADC$ are similar. 2
- (ii) Prove that $AB \times CD = AC \times BE$. 1
- (iii) Hence by proving that another pair of triangles are similar, deduce that $AB \times CD + AD \times BC = AC \times BD$. 4

THE END

QUESTION 1

(a) (i) $\frac{a}{1+i} + \frac{b}{1+2i} = 1$

$a(1+2i) + b(1+i) = (1+i)(1+2i)$

$(a+b) + i(2a+b) = -1 + 3i$

$a+b = -1 \quad \text{--- (1)}$

$2a+b = 3 \quad \text{--- (2)}$

(2) - (1) $a = 4$

from (1) $4+b = -1$

$b = -5$

$\therefore a = 4, b = -5$

(b) (i) $z^2 = r^2 \cos 2\theta$

$|z^2| = r^2, \arg(z^2) = 2\theta \quad \text{or} \quad |z^2| = |z|^2, \arg(z^2) = 2\arg z$
 $= r^2, \quad = 2\theta$

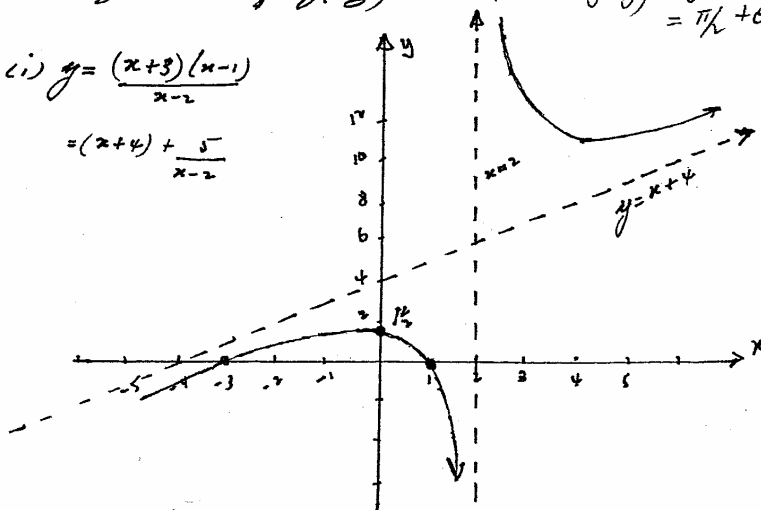
(ii) $\frac{1}{z} = z^{-1} = r^{-1} \cos(-\theta) \quad \text{or} \quad \frac{1}{|z|} = \frac{1}{|z|}, \arg\left(\frac{1}{z}\right) = \arg(1) - \arg z$
 $\frac{1}{|z|} = \frac{1}{r}, \arg\left(\frac{1}{z}\right) = -\theta \quad = \frac{1}{r}, \quad = -\theta$

(iii) $iz = \cos \pi/2 \cdot r \cos \theta \quad \text{or} \quad |iz| = |i||z|$
 $= r \cos(\theta + \pi/2) \quad = r$

$|iz| = r, \arg(iz) = \theta + \pi/2 \quad \arg(iz) = \arg(i) + \arg z$
 $= \pi/2 + \theta$

(c) (i) $y = \frac{(x+3)(x-1)}{x-2}$

$= (x+4) + \frac{5}{x-2}$



(2)

Q1(c)(ii) $A = \int_{-3}^1 \frac{x+4+5}{x-2} dx$

$$\begin{aligned}
 &= \left[\frac{1}{2}x^2 + 4x + 5\ln(2-x) \right]_{-3}^1 \\
 &= \left(\frac{1}{2} + 4 + 5\ln 1 \right) - \left(\frac{9}{2} - 12 + 5\ln 5 \right) \\
 &= 12 - 5\ln 5 \\
 \text{Area} &= 12 - 5\ln 5 \quad u^2
 \end{aligned}$$

QUESTION 2

(a)(i) $\left[\frac{1}{4} \sin^4 \theta \right]_0^{\pi/6} = \frac{1}{4} (\sin^4 \pi/6 - \sin^4 0)$
 $= \frac{1}{4} \left(\frac{1}{2}^4 - 0 \right)$
 $= \frac{1}{64}$

(ii) $u^2 = x \quad x=0, u=0$
 $2u du = dx \quad x=3, u=\sqrt{3}$

$$\begin{aligned}
 \int_0^3 \frac{\sqrt{x}}{1+x} dx &= \int_0^{\sqrt{3}} \frac{u}{1+u^2} \cdot 2u du \\
 &= 2 \int_0^{\sqrt{3}} \frac{u^2}{1+u^2} du \\
 &= 2 \int_0^{\sqrt{3}} \left(1 - \frac{1}{1+u^2} \right) du \\
 &= 2 \left[u - \tan^{-1} u \right]_0^{\sqrt{3}} \\
 &= 2 \left\{ (\sqrt{3} - \tan^{-1} \sqrt{3}) - (0 - \tan^{-1} 0) \right\} \\
 &= 2(\sqrt{3} - \pi/6)
 \end{aligned}$$

(iii) let $t = \tan \theta/2$, $\cos \theta = \frac{1-t^2}{1+t^2}$ $d\theta = \frac{2dt}{1+t^2}$
 $\theta=0 \quad t=0, \quad \theta=\pi/2, \quad t=1$

$$\int_0^1 \frac{1}{5+3\left(\frac{1-t^2}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2} = 2 \int_0^1 \frac{dt}{5(1+t^2)+3(1-t^2)}$$

(3)

Q2(a)(iii)

$$\begin{aligned}
 &= 2 \int_0^1 \frac{dt}{8+2t^2} \\
 &= \int_0^1 \frac{dt}{4+t^2} \\
 &= \frac{1}{2} \left[\tan^{-1} \frac{t}{2} \right]_0^1 \\
 &= \frac{1}{2} \tan^{-1} \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)(i)} \quad I_n &= \int_0^{\pi/2} x^n \cdot \frac{d}{dx} (-\cos x) dx \\
 &= \left[-x^n \cos x \right]_0^{\pi/2} - \int_0^{\pi/2} -\cos x \cdot n x^{n-1} dx \\
 &= 0 + n \int_0^{\pi/2} x^{n-1} \cos x dx \\
 &= n \int_0^{\pi/2} x^{n-1} \cdot \frac{d}{dx} (\sin x) dx \\
 &= n \left[x^{n-1} \sin x \right]_0^{\pi/2} - n \int_0^{\pi/2} \sin x \cdot (n-1) x^{n-2} dx \\
 &= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) \int_0^{\pi/2} x^{n-2} \sin x dx \\
 &= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}
 \end{aligned}$$

$$I_n + n(n-1) I_{n-2} = n \left(\frac{\pi}{2} \right)^{n-1}$$

$$\text{(ii)} \quad \int_0^{\pi/2} x^4 \sin x dx = I_4$$

$$I_4 = -4(3) I_2 + 4 \left(\frac{\pi}{2} \right)^3$$

$$I_2 = -2(1) I_0 + 2 \left(\frac{\pi}{2} \right)^1$$

$$\begin{aligned}
 I_0 &= \int_0^{\pi/2} \sin x dx \\
 &= 1
 \end{aligned}$$

$$\therefore I_2 = -2(1) + \pi$$

④.

Q2(b)(ii)

$$I_4 = -12(-2 + \pi) + \frac{\pi^3}{2}$$

$$= 24 - 12\pi + \frac{\pi^3}{2}$$

QUESTION 3

(a)(i) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

$a=5 \quad b=3$

$b^2 = a^2(1 - e^2)$

$x\text{-int } (5, 0)$

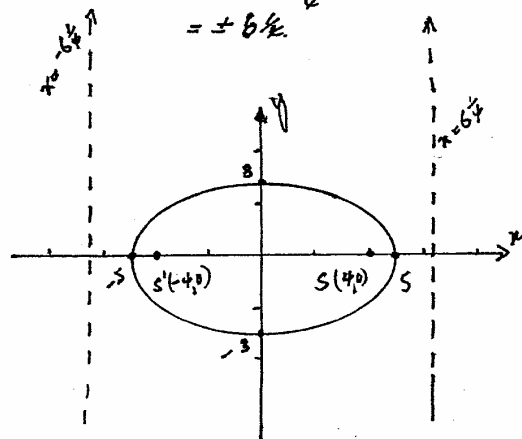
$9 = 25(1 - e^2)$

$y\text{-int } (0, \pm 3)$

$e = 4/5$

(ii) Foci $(\pm ae, 0) = (\pm 5 \cdot \frac{4}{5}, 0)$
 $= (\pm 4, 0)$

(iii) vertices $x = \pm a/c$
 $= \pm 5 \cdot \frac{5}{4}$
 $= \pm 6\frac{1}{4}$



(b) $x^2 - y^2 = c^2$
 $2x - 2y \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = \frac{x}{y} = m_1$

$xy = c^2$
 $y + x \frac{dy}{dx} = 0$

Q3(b)

$$\frac{dy}{dx} = -y/x = m_2$$

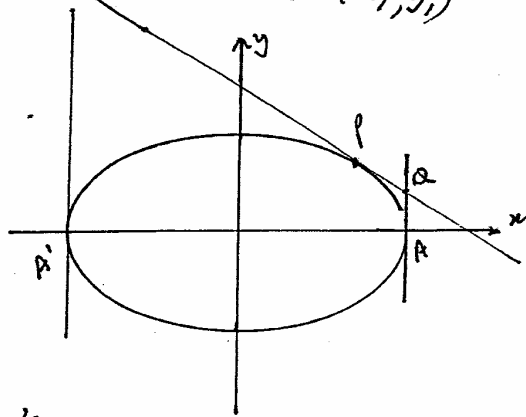
(5)

$$\text{at } (x_1, y_1) \quad m_1 = \frac{x_1}{y_1} \quad \& \quad m_2 = -\frac{y_1}{x_1}$$

$$m_1 m_2 = \frac{x_1}{y_1} \cdot -\frac{y_1}{x_1} = -1$$

$\therefore$ curves are $\perp$ at (x_1, y_1)

(c)



$$(i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{2x}{a^2} \cdot \frac{b^2}{2y} \\ &= -\frac{b^2 x}{a^2 y} \end{aligned}$$

$$\begin{aligned} \text{at P} \quad \frac{dy}{dx} &= -\frac{b^2}{a^2} \cdot \frac{a \cos \theta}{b \sin \theta} \\ &= -\frac{b \cos \theta}{a \sin \theta} \end{aligned}$$

$$\text{tangent } y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\begin{aligned} a \sin \theta y - ab \sin^2 \theta &= -b \cos \theta x + ab \cos^2 \theta \\ b \cos \theta x + a \sin \theta y &= ab(\sin^2 \theta + \cos^2 \theta) \\ b \cos \theta x + a \sin \theta y &= ab \end{aligned}$$

Q3(c)(ii) $b \cos \theta x + a \sin \theta y = ab$

⑥.

at A, $x = a \therefore ab \cos \theta + a \sin \theta y = ab$

$$y = \frac{ab(1 - \cos \theta)}{a \sin \theta}$$

$$= \frac{b(1 - \cos \theta)}{\sin \theta}$$

Q is $\left(a, \frac{b(1 - \cos \theta)}{\sin \theta}\right)$

at A', $x = -a \therefore -ab \cos \theta + a \sin \theta y = ab$

$$y = \frac{ab(1 + \cos \theta)}{a \sin \theta}$$

$\therefore Q' \text{ is } \left(-a, \frac{b(1 + \cos \theta)}{\sin \theta}\right)$

(iii) $AQ \cdot A'Q' = \frac{b(1 - \cos \theta)}{\sin \theta} \cdot \frac{b(1 + \cos \theta)}{\sin \theta}$

$$= \frac{b^2(1 - \cos^2 \theta)}{\sin^2 \theta}$$

$$= \frac{b^2(\sin^2 \theta)}{\sin^2 \theta}$$

$$= b^2$$

$\therefore AQ \cdot A'Q'$ is independent of θ .

QUESTION 4.

(a) $\frac{d}{dx} \left[(6x - x^2)^{\frac{3}{2}} + \frac{b}{2} \cos^{-1} \left(\frac{2x - b}{b} \right) \right]$

$$= \frac{3}{2} (6x - x^2)^{-\frac{1}{2}} \cdot (6 - 2x) + \frac{b}{2} \cdot \frac{-\frac{2}{b}}{\sqrt{1 - \left(\frac{2x - b}{b} \right)^2}}$$

$$= \frac{6 - 2x}{2 \sqrt{6x - x^2}} - \frac{1}{\sqrt{\frac{b^2 - (2x - b)^2}{b^2}}}$$

$$= \frac{6 - 2x}{2 \sqrt{6x - x^2}} - \frac{b}{\sqrt{b^2 - 4x^2 + 4bx - b^2}}$$

Q4(a)

⑦

$$= \frac{b-2x}{2\sqrt{bx-x^2}} - \frac{b}{\sqrt{4bx-4x^2}}$$

$$= \frac{b-2x-b}{2\sqrt{bx-x^2}}$$

$$= \frac{-x}{\sqrt{bx-x^2}}$$

$$= -\sqrt{\frac{x^2}{bx-x^2}} \quad \text{for } x \geq 0$$

$$= -\sqrt{\frac{x}{b-x}}$$

$$(b)(i) m \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -\mu m x^{-2} \quad (x'' < 0)$$

$$\frac{1}{2} v^2 = \mu x^{-1} + C$$

$$t=0, x=b, v=0$$

$$0 = \frac{\mu}{b} + C$$

$$C = -\frac{\mu}{b}$$

$$\frac{1}{2} v^2 = \frac{\mu}{x} - \frac{\mu}{b}$$

$$v^2 = 2\mu \left(\frac{b-x}{bx} \right)$$

(ii) Since $v \neq 0$ for $x > 0$ then particle does not stop.

$\therefore$ motion is in same dir. as initial motion
& since $x'' < 0$ when $x=b$ then direction of motion
is towards origin for $x > 0$

$$\therefore v = -\sqrt{\frac{2\mu}{bx} (b-x)}$$

$$\frac{dx}{dt} = -\sqrt{\frac{2\mu}{b}} \cdot \sqrt{\frac{b-x}{x}}$$

⑧

$$Q_4(b)(ii) \quad \frac{dt}{dx} = -\sqrt{\frac{b}{2\mu}} \cdot \sqrt{\frac{x}{b-x}}$$

$$t = \int_b^{b/2} -\sqrt{\frac{b}{2\mu}} \cdot \sqrt{\frac{x}{b-x}} dx$$

$$= \sqrt{\frac{b}{2\mu}} \left[\sqrt{bx-x^2} + \frac{b}{2} \cos^{-1}\left(\frac{2x-b}{b}\right) \right]_b^{b/2}$$

$$= \sqrt{\frac{b}{2\mu}} \left\{ \left(\sqrt{\frac{b^2}{4} - \frac{b^2}{4}} + \frac{b}{2} \cos^{-1}(0) \right) - \left(\sqrt{b^2 - b^2} + \frac{b}{2} \cos^{-1}(1) \right) \right\}$$

$$= \sqrt{\frac{b}{2\mu}} \left\{ \sqrt{\frac{b^2}{4}} + \frac{b \cdot \pi}{2} - 0 \right\}$$

$$= \sqrt{\frac{b}{2\mu}} \left(\frac{b}{2} + \frac{b\pi}{2} \right)$$

$$\text{time} = \frac{b(2+\pi)}{4} \cdot \sqrt{\frac{b}{2\mu}}$$

(c) when $n=2$, $(n+1)^2 - 2n - 1 = x^2 + 2n + 1 - 2n - 1 = x^2$

$\therefore$ true for $n=2$

Assume true for $n \leq k$ (k an integer)

$$\text{i.e. } (n+1)^k - kn - 1 = x^2 P(x)$$

To prove true for $n=k+1$

$$\text{i.e. } (n+1)^{k+1} - (k+1)n - 1 = x^2 Q(x)$$

$$\begin{aligned} \text{LHS} &= (n+1)^{k+1} - (k+1)n - 1 \\ &= (n+1) \cdot (n+1)^k - (k+1)n - 1 \\ &= (n+1) [x^2 P(x) + kn + 1] - (k+1)n - 1 \\ &= x^3 P(x) + kn^2 + n + x^2 P(x) + kn + 1 - kn - n - 1 \\ &= x^3 P(x) + x^2 k + x^2 P(x) \\ &= x^2 (x P(x) + P(x) + k) \\ &= x^2 Q(x) \end{aligned}$$

$\therefore$ if true for $n=k$ then true for $n=k+1$

$\therefore$ since true for $n=2$ then true for all integers $n \geq 2$.

QUESTIONS

(9).

(a) (i) $x = 2 \cos \theta$

$x=0, \theta=0$

$dx = -2 \sin \theta d\theta$

$x=2, \cos \theta = 1$
 $\theta = \pi/2$

$$\begin{aligned} \int_0^2 (4-x^2)^{3/2} dx &= \int_0^{\pi/2} (4-4\sin^2 \theta)^{3/2} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\pi/2} 8(1-\sin^2 \theta)^{3/2} \cdot 2 \cos \theta d\theta \\ &= 16 \int_0^{\pi/2} \cos^3 \theta \cdot \cos \theta d\theta \\ &= 16 \int_0^{\pi/2} \cos^4 \theta d\theta \end{aligned}$$

(ii) $AB = y = 4-x^2$

$BC = 2^2 - x^2$

$BC = \sqrt{4-x^2} \quad (BC > 0)$

Area $\triangle ABC = \frac{1}{2} (4-x^2) \sqrt{4-x^2}$
 $= \frac{1}{2} (4-x^2)^{3/2}$

$\Delta V = \text{area cross-section} \times \text{thickness}$
 $\approx \frac{1}{2} (4-x^2)^{3/2} \cdot \Delta x$

$V = \lim_{\Delta x \rightarrow 0} \sum_{-2}^2 \frac{1}{2} (4-x^2)^{3/2} \Delta x$
 $= \frac{1}{2} \int_{-2}^2 (4-x^2)^{3/2} dx$

$V = \int_0^2 (4-x^2)^{3/2} dx$ since function is even.

(iii) $V = \int_0^2 (4-x^2)^{3/2} dx$

$= 16 \int_0^{\pi/2} \cos^4 \theta d\theta$

$= 16 \int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2} \right)^2 d\theta$

$= 4 \int_0^{\pi/2} 1 + 2\cos 2\theta + \cos^2 2\theta d\theta$

$\cos 2\theta = 2\cos^2 \theta - 1$

$\cos^2 \theta = \frac{1+\cos 2\theta}{2}$

Q5(a)(iii) $V = 4 \int_0^{\pi/4} 1 + 2\cos 2\theta + \frac{1 + \cos 4\theta}{2} d\theta$

$$= 2 \int_0^{\pi/4} 3 + 4\cos 2\theta + \cos 4\theta d\theta$$

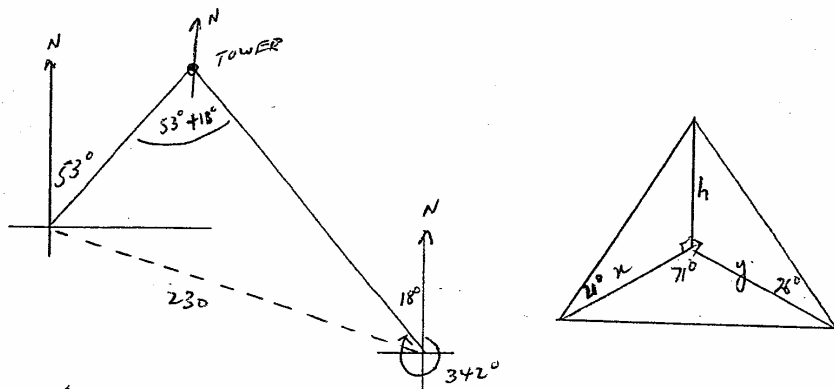
$$= 2 \left[3\theta + 2\sin 2\theta + \frac{1}{4}\sin 4\theta \right]_0^{\pi/4}$$

$$= 2 \left\{ \frac{3\pi}{4} + 2\sin \pi + \frac{1}{4}\sin 2\pi - 0 \right\}$$

$$= 3\pi$$

Volume = $3\pi \text{ u}^3$

(b)



$$\frac{h}{x} = \tan 21^\circ$$

$$x = \frac{h}{\tan 21^\circ}$$

$$= h \cot 21^\circ$$

$$\frac{h}{y} = \tan 26^\circ$$

$$y = h \cot 26^\circ$$

$$230^2 = (h \cot 21^\circ)^2 + (h \cot 26^\circ)^2 - 2(h \cot 21^\circ)(h \cot 26^\circ) \cos 71^\circ$$

$$h = \frac{230}{\sqrt{\cot^2 21^\circ + \cot^2 26^\circ - 2 \cot 21^\circ \cot 26^\circ \cos 71^\circ}}$$

$$= 83.91 \text{ m}$$

height = 84 m (to nearest m)

QUESTION 6

11

(a) (i) $y = c^2/x$
 $y' = -c^2/x^2$

at C, $y' = \frac{-c^2}{c^2 t^2}$
 $= -\frac{1}{t^2}$

Tangent: $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$
 $t^2 y - ct = -x + ct$
 $x + t^2 y = 2ct$

(ii) Normal: $tx^2 - y = R$

at C: $t^2(ct) - \frac{c}{t} = k$

$k = \frac{c(t^4 - 1)}{t}$

Normal: $t^2 x - y = \frac{c(t^4 - 1)}{t}$

$t^3 x - y = c(t^4 - 1)$

(iii) at F, $y = 0$ $\therefore x = 2ct$ $F(2ct, 0)$

at G, $x = 0$ $t^2 y = 2ct$

$y = \frac{2c}{t}$ $(t \neq 0)$ $G(0, \frac{2c}{t})$

for H $y = x$ — (1)

$t^3 x - ty = c(t^4 - 1)$ — (2)

Sub (1) into (2)

$t^3 x - tx = c(t^4 - 1)$

$x = \frac{c(t^2 - 1)(t^2 + 1)}{t(t^2 - 1)} = \frac{c(t^2 + 1)}{t}$ $(t \neq 0)$

$\therefore H \rightarrow \left[\frac{c(t^2 + 1)}{t}, \frac{c(t^2 + 1)}{t} \right]$

(12)

$$\begin{aligned}
 m(FH) &= \frac{\frac{c(t^2+1)}{t} - 0}{\frac{c(t^2+1) - 2ct}{t}} \\
 &= \frac{c(t^2+1)}{c(t^2+1) - 2ct} \\
 &= \frac{c(t^2+1)}{c(1-t^2)} \\
 &= \frac{1+t^2}{1-t^2}
 \end{aligned}$$

$$\begin{aligned}
 m(GH) &= \frac{\frac{c(t^2+1)}{t} - \frac{2c}{t}}{\frac{c(t^2+1)}{t}} \\
 &= \frac{c(t^2+1) - 2c}{c(t^2+1)} \\
 &= \frac{t^2-1}{t^2+1}
 \end{aligned}$$

$$\begin{aligned}
 m(FH) \cdot m(GH) &= \frac{1+t^2}{1-t^2} \cdot \frac{t^2-1}{t^2+1} \\
 &= \frac{t^2-1}{1-t^2} \\
 &= \frac{t^2-1}{-(t^2-1)} \\
 &= -1
 \end{aligned}$$

$\therefore FH \perp GH$ (prod. slopes = -1)

Q6(b) $V = 2\pi \int_1^e x \frac{\ln x}{\sqrt{x}} dx$

(13)

$$= 2\pi \int_1^e \sqrt{x} \ln x dx$$

$$= 2\pi \int_1^e \ln x \cdot \frac{d}{dx} \left(\frac{2}{3} x^{3/2} \right) dx$$

$$= 2\pi \left\{ \left[\frac{2}{3} x^{3/2} \ln x \right]_1^e - \int_1^e \frac{2}{3} x^{3/2} \cdot \frac{1}{x} dx \right\}$$

$$= 2\pi \left\{ \left(\frac{2}{3} e^{3/2} \ln e - 0 \right) - \frac{2}{3} \int_1^e x^{1/2} dx \right\}$$

$$= 2\pi \left\{ \frac{2}{3} e^{3/2} - \frac{2}{3} \left[\frac{2}{3} x^{3/2} \right]_1^e \right\}$$

$$= 2\pi \left\{ \frac{2}{3} e^{3/2} - \frac{4}{9} (e^{3/2} - 1) \right\}$$

$$Vol. = \frac{4\pi}{9} (e^{3/2} + 2) u^3$$

QUESTION 7

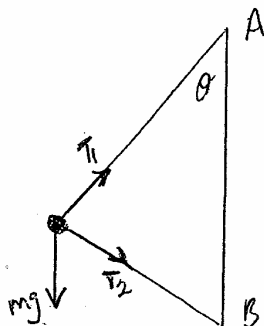
$$(a)(i) N^{\circ} \text{ ways} = \frac{{}^{12}C_4 \cdot {}^8C_4 \cdot {}^4C_4}{3!}$$

$$= 5775$$

$$(ii) N^{\circ} \text{ ways} = {}^{10}C_3 \cdot {}^7C_3 \cdot {}^4C_4$$

$$= 4200$$

(b)(i)



Q 7(b) (iii)

(14)

Vertically: $T_1 \cos \theta = mg + T_2 \sin \theta$ — (1)

Horizontally: $T_1 \sin \theta + T_2 \cos \theta = m r \omega^2$ — (2)

(1) $\times \cos \theta$

$$T_1 \cos^2 \theta - T_2 \sin \theta \cos \theta = mg \cos \theta \quad (3)$$

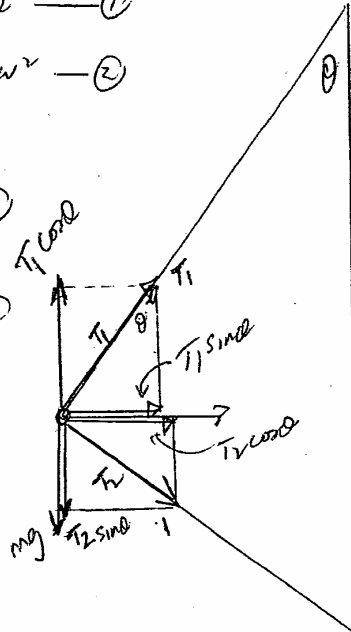
(2) $\times \sin \theta$

$$T_1 \sin^2 \theta + T_2 \sin \theta \cos \theta = m r \omega^2 \sin \theta \quad (4)$$

(3) + (4)

$$T_1 (\sin^2 \theta + \cos^2 \theta) = mg \cos \theta + m r \omega^2 \sin \theta$$

$$T_1 = mg \cos \theta + m r \omega^2 \sin \theta$$



So (1) $T_2 \sin \theta = T_1 \cos \theta - mg$

$$= mg \cos \theta + m r \omega^2 \sin \theta \cos \theta - mg$$

$$= mg (1 - \sin^2 \theta) + m r \omega^2 \sin \theta \cos \theta - mg$$

$$= mg - mg \sin^2 \theta + m r \omega^2 \sin \theta \cos \theta - mg$$

$$= m r \omega^2 \sin \theta \cos \theta - mg \sin^2 \theta$$

$$T_2 = m r \omega^2 \sin \theta \cos \theta - mg \sin^2 \theta$$

(iii) $\sin \theta = 0.6$

$\cos \theta = 0.8$ $AP = 0.8$

$$r = 0.8 - \sin \theta$$

$$= 0.8 - 0.6$$

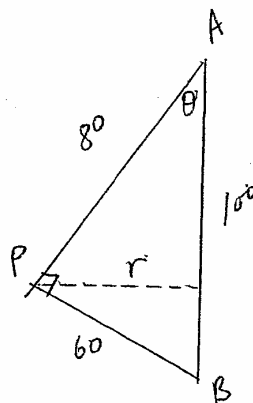
$$= 0.2$$

if $T_2 > 0$

$$m(r \omega^2 \cos \theta - g \sin \theta) > 0$$

$$r \omega^2 \cos \theta > g \sin \theta$$

$$\omega^2 > \frac{g \sin \theta}{r \cos \theta}$$



Q7(b)(iii) (cont)

(15)

$$\omega^2 > \frac{g \times 0.6}{0.48 \times 0.8}$$

$$\omega^2 > \frac{25g}{16}$$

(iv) If free to move $T_1 = T_2$

$$m(r\omega^2 \sin \theta + g \cos \theta) = m(r\omega^2 \cos \theta - g \sin \theta)$$

$$r\omega^2 \sin \theta + g \cos \theta = r\omega^2 \cos \theta - g \sin \theta$$

$$g(\cos \theta + \sin \theta) = r\omega^2(\cos \theta - \sin \theta)$$

$$\omega^2 = \frac{g(\cos \theta + \sin \theta)}{r(\cos \theta - \sin \theta)}$$

$$= \frac{g(0.8 + 0.6)}{0.48(0.8 - 0.6)}$$

$$= \frac{175g}{12}$$

QUESTION 8

$$(a)(i) \tan 4x = \frac{2 \tan 2x}{1 - \tan^2 2x}$$

$$= \frac{2 \left(\frac{2t}{1-t^2} \right)}{1 - \left(\frac{2t}{1-t^2} \right)^2}$$

where $t = \tan x$

$$= \frac{\frac{4t}{1-t^2}}{\frac{(1-t^2)^2 - (2t)^2}{(1-t^2)^2}}$$

$$Q8(a) (i) \quad \tan x = \frac{4t}{1-t^2} \times \frac{(1-t^2)^2}{1-2t^2+t^2-4t^2}$$

(18)

$$= \frac{4t(1-t^2)}{1-6t^2+t^4}$$

$$(ii) \quad \tan x \tan 4x = 1$$

$$\frac{t(4t)(1-t^2)}{t^4-6t^2+1} = 1$$

$$4t^2(1-t^2) = t^4-6t^2+1$$

$$4t^2-4t^4 = t^4-6t^2+1$$

$$5t^4-10t^2+1=0$$

$$(iii) \quad x=18^\circ, \quad \tan x \tan 4x = \tan 18^\circ \tan 72^\circ \\ = \tan 18^\circ \cot 18^\circ \\ = 1$$

$$x=54^\circ, \quad \tan x \tan 4x = \tan 54^\circ \tan 216^\circ \\ = \tan 54^\circ \tan 36^\circ \\ = \tan 54^\circ \cot 54^\circ \\ = 1.$$

$$(iv) \quad 5t^4-10t^2+1=0$$

$$t^2 = \frac{10 \pm \sqrt{100-20}}{10}$$

$$= \frac{5 \pm 2\sqrt{5}}{5}$$

Now the roots of the above equation are $\pm \tan 18^\circ$ and $\pm \tan 54^\circ$ and since $\tan 18^\circ < \tan 54^\circ$ then $\tan 54^\circ = \sqrt{\frac{5+2\sqrt{5}}{5}}$

QUESTION 8 (b)

(17)

(i) In $\triangle ABE$ & $\triangle ACD$

$$\hat{ABE} = \hat{ACD} \text{ (angles at circumference)}$$

$$\hat{EAB} = \hat{DAC} \text{ (on same arc AD)} \\ \text{(EAB} + \theta \text{)}$$

$\therefore \triangle ABE \parallel \triangle ACD$ (equiangular)

(ii) $\frac{AB}{AC} = \frac{AE}{AD} = \frac{BE}{DC}$ (ratio of corresponding sides in similar triangles)

$$AB \cdot CD = AC \cdot BE$$

(iii) In $\triangle ABC$ & $\triangle AED$

$$\hat{BAC} = \hat{DAE} \text{ (both } \theta \text{)}$$

$$\hat{BCA} = \hat{EDA} \text{ (angles at circumference on same arc AB)}$$

$\therefore \triangle ABC \parallel \triangle ADE$ (equiangular)

$$\frac{BC}{DE} = \frac{AC}{AD} \left(= \frac{AB}{AE} \right) \text{ (ratio of corresponding sides in similar triangles)}$$

$$BC \cdot AD = DE \cdot AC$$

$$BC \cdot AD + AB \cdot CD = DE \cdot AC + BE \cdot AC \quad \text{(from (ii) + (iii))} \\ = AC(DE + BE)$$

$$AB \cdot CD + BC \cdot AD = AC \cdot BD$$

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